INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

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Statistical Models
Week 10: Markov Chain Monte Carlo: the Gibbs sampler

comments

> ! There are no lectures on Wednesday, November 5th.
>
> Instead, we will have a longer lecture on Friday, November 7th lectures within 10:15-13:45, with a lunch break of at least half an hour. Friday lecture will be in Aud D1.
> In course week 10, there will only be lectures on Friday, November 7th. Next week, lectures on Wednesday are extended. Since the exercise class is moved from Tuesday to Wednesday you will have a free Tuesday early morning.

This week's only lecture concentrates on the *Gibbs sampler*, a method that allows to generate random samples from the posterior distribution a multidimensional parameter. In essence it consists in updating one element of the multidimensional parameter at a time.

The other very important MCMC method is the *Metropolis-Hastings algorithm*, that enables us to simulate from a distribution that is known up to a constant. If time allows, we will start with the Metropolis algorithm in the end of Friday's lecture.

In order to understand the algorithms, you will implement them yourselves in R. Later in this course you will use software "off the shelf" to run MCMC. You will learn to use JAGS and the R package `rjags` to analyze simple Bayesian models.

Install JAGS

In the coming weeks, we will use the R package `rjags` for Bayesian analysis. `rjags` is an interface to the software `JAGS` (Just Another Gibbs Sampler) that you need to install separately on your computer (not from within R). Please install it from `sourceforge` (link).

Relevant Readings

- Notes on Markov chains with finite and continuous state space,
- Gibbs sampler: Reich & Ghosh, Section 3.2, 3.2.1.

Lecture Material

- Link to onenotes on Google drive
- Gibbs sampler for bivariate normal distribution
- Listing 3.2 from Reich & Ghosh for experimentation
- Proof of the Gibbs sampler

Exercises

tba